

文献阅读笔记

灾害救援模块化空地车队的集成系统设计

文献信息 A. C. Gärtner, D. Ferriero, A. E. Bayrak, P. Y. Papalambros. "Integrated System Design of a Modular, Autonomous, Aerial and Ground Vehicle Fleet for Disaster Relief Missions—A Case Study." *International Design Conference—DESIGN 2018*, 2018. DOI: 10.21278/idc.2018.0477。

一句话概括 论文证明：模块划分不能只依据部件之间的静态交互，还要把 UAV/UGV 车队放入随时间演化的救灾任务中评价；模块化与自主协同的真正收益，来自 UGV 自动更换 UAV 电池等车队级行为。

场景与需求

作者构造了类似 2016 年意大利 Amatrice 地震的场景：约 4 km² 城区、15,000 人口、50 人失踪、道路与通信受损、建筑倒塌且存在燃气泄漏。车队需要完成五类任务：灾情记录、搜救、物资/废墟运输、通信恢复和气体检测。

由此得到的关键设计约束包括：单车至少工作 30 min、总体任务持续 48 h、电池可更换、UGV 可无人介入地为 UAV 换电；同时要求便于运输、轻量、耐候、易修复，并支持 GPS 拒止环境中的自主导航。

八阶段设计流程

作者把“定义—发散—过渡—收敛—原型”的简化流程细化为八阶段：问题定义；概念设计的发散、过渡、收敛；实体设计的发散、过渡、收敛；最终原型。概念层与实体层都允许迭代，这使模块边界、任务性能和结构实现可以互相反馈。

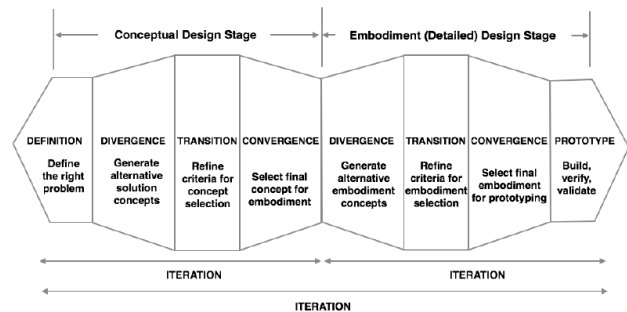


Figure 1. The design process

4. Conceptual design

Conceptual design involves functional development of UAV and UGV concepts in view of the desired design attributes. Both ideation techniques and computational methods are employed to create multiple concept alternatives. A concept selection method such as a Pugh Chart is used to select a concept for further development.

4.1. Concept generation

Following the above process model, we performed two brainstorming sessions with a group of eight design researchers at the University of Michigan. The first session was structured as an open-ended session with researchers working in pairs and developing mind maps. A second group discussion session was conducted converging to two conceptual design solutions for the overall fleet. In addition to brainstorming, we used a DSM-based approach to create additional design alternatives. We identified 17 physical elements in the UAVs and UGVs combined, created the DSM based on interactions among the (functional or physical) elements of the fleet design, and defined multiple objectives such as minimizing the size of the modules and minimizing the interactions among modules to draw boundaries in the DSM that define the modules (Eppinger and Browning, 2012). We omit the DSM presentation here for brevity. We then formulated an optimization problem to minimize the two objectives namely, sum of module sizes and the number of required interactions/interfaces among modules, with respect to the clustering the components. Solving the multiobjective optimization problem, we obtained 30 more concepts. Here we discuss six representative designs. Concept 1 generated in the brainstorming session places a Wi-Fi module on a blimp due to the stationary nature of the mission. This concept assumes separate platforms with essential electronics and drivetrain components for UAVs and UGVs. Since gas sensing is required for all vehicles, the platforms also contain gas sensors. Battery and all other mission-related sensors are defined as separate modules in this concept. Concept 2 is similar to Concept 1 except it combines sensors required by the same mission into a single module. It defines a search module with thermal camera, radars, documentation module with HD camera and another module with 3D scanner, and a transportation module. Concepts 3-6 are created by the DSM method. Concept 3 is a non-modular, multifunctional vehicle that can perform both aerial and ground missions with all the necessary components available on the vehicle. Concept 6 is at the opposite end of the spectrum with almost all elements separately defined as modules including the Wi-Fi device. It only combines motors, ECS and battery as a module. Concept 4 has separate platforms for UAVs and UGVs with essential electronics and 3D scanners, payload attachment and thermal cameras. Drivetrain and other mission sensors are located in separate modules. Concept 5 has motors and rotors for UAVs, and rolling gear for UGVs, the

图：原文第 4 页，展示扩展后的设计流程，以及从需求进入概念方案的路径。

概念生成与车队仿真

研究用两条路线生成方案：8 名设计研究者分组头脑风暴；以及基于 DSM（设计结构矩阵）对 17 个物理元素做多目标聚类，目标是减小模块尺寸和模块间接口数。DSM 路线产生 30 个概念，论文选出 6 个代表方案，覆盖从高度集成到高度模块化的光谱。

随后建立时间步进车队仿真。失踪人员、倒塌建筑和燃气泄漏位置随机，车队完成搜索、救援和记录任务。每个概念执行 100 次蒙特卡洛模拟，评价指标包括：总冗余、车辆数量、单车冗余、重构次数、找到 50 名失踪人员的总时间和充电次数。

一个关键观察是：不可换电 UAV 频繁返回基地充电，显著浪费任务时间；方案若减少不必要的模块重构，也会获得更好的车队性能。

方案选择结果

作者加入适应性、维护性、运输性、升级性、开发工作量、制造和重量等工程判断指标，再用加权表选型。最终选择的是头脑风暴产生的 Concept 2，总分 419；非模块化的 Concept 3 在车队仿真中排名第二，而 DSM 生成的方案整体并未占优。

这并不意味着 DSM 无用，而是说明 DSM 只表达预先定义的部件交互，难以覆盖任务触发、时间成本、车辆调度和自主换电等运行层属性。

UAVs and UGVs search for missing people following predefined trajectories. Figure 2 shows how a UAV without battery replacement capability scans the area and comes back for battery recharge. We see from the simulation results that if the automated battery replacement is not an option, UAVs spend a lot of time for returning back to the base station for recharging. Therefore, it is essential to design the UAV and UGV teaming as a physical system and a control system to maximize fleet performance. Vehicles also return to the base station for reconfigurations. Concepts that minimize the number of necessary reconfigurations perform better in the fleet simulation.

4.3. Concept selection

Following the simulation results, we defined further criteria to inform the design selection process. We used the chart shown in Table 2 to compare the six concepts evaluated on a scale of 1-5. We divided the performance in the fleet simulation further into sub-criteria based on the level of information provided by the simulation model. Additional criteria shown in Table 2 in bold cannot easily be evaluated by a simulation and hence require engineering judgement at the early design stages due to lack of information related to embodiment. We assign subjective weighting based on the pairwise comparison of each metric on a scale of three levels (more important, equally important, less important) using engineering judgement. Modular design makes a significant difference at this point. Modular systems are easier to maintain and upgrade, and enable lightweight design. Also, the manufacturing process may benefit from choosing modular solutions. However, the development effort for modular design could be higher compared to integrated solutions. We selected Concept 2 created by brainstorming. Interestingly, the designs created by the DSM method (Concepts 3-6) were judged inferior, likely because modularity decisions based on predefined interactions among components do not yield optimal results when fleet-level operational metrics are considered. The non-modular design in Concept 3 performs second best in the fleet simulation only surpassed by Concept 2. To fully benefit from modularity, the design concepts must be evaluated at the fleet level considering factors beyond those in the DSM.

Table 2. Comparison of concept alternatives							
Criteria	Weight	C1	C2	C3	C4	C5	C6
Fleet simulation performance	25	2,3	4,2	3,9	1,5	2,7	3,5
Total redundancy	27,5	4	5	3	1	2	3
Number of vehicles	27,5	1	4	5	2	2	4
Reconfigurations	20	1	4	5	1	4	4
Simulation time	12,5	2	4	3	2	4	4
Redundancy/ # of vehicles	7,5	5	4	2	1	2	2
Number of recharges	5	2	3	3	3	2	2
Adaptability	2	3	3	4	3	3	3
Maintainability	20	5	4	2	3	3	3
Transportability	20	3	5	4	1	2	4
Upgradability	9	4	4	2	3	3	3
Development effort	11	2	3	4	3	3	3
Manufacturing	2	3	3	2	3	3	3
Vehicle weight	13	5	4	1	3	4	3
Sum	510	352	419	304	229	290	338

5. Embodiment design: UAV

Next, we developed the embodiment for the modular UAV design (modular UGV is a future task). Following the process model again, we generated multiple alternatives, refined the selection criteria and made a final choice among the alternatives.

图：原文第 6 页。Concept 2 的综合评价最高；表格也展示了性能与工程判断如何合并。

UAV 实体设计

- 构型：在四、六、八旋翼中选择六旋翼，平衡单电机故障鲁棒性、载荷、尺寸和机动性。
- 动力：通过 eCalc 迭代电机、桨、ESC、机架和电池；最终示例选用 DJI Snail 2305 电机、5048S 三叶桨和 7.4 V/3200 mAh 电池。

- 控制：Pixhawk 2.1 负责飞控，Raspberry Pi 负责自主功能；传感器包括 LiDAR Lite、高清相机、热像仪与 MQ-2 气体传感器。
- 模块接口：从螺钉、卡环、卡口、快接等候选中选择锁扣式接口。
- 二代机架：由六边形三层开放结构，迭代为圆形封闭抗扭结构；锁扣从 6 个减至 3 个，并根据有限元高应力区增加支撑。
- 换电：UAV 直接落在电池壳体上，电池沿导轨抽换，为 UGV 自动更换留出接口。



Figure 6. Example of an early prototype built for design exploration



Figure 7. Prototype of the second-generation design embodiment

We built the final prototype with 3D printing for the UAV frame using polycarbonate. We printed the main plate, main body, module holder, LiDAR module, as well as motor mounts, arm tubes, and arm connectors. This prototype, shown in Figure 7, demonstrates the feasibility of the modular design, mechanical interfaces and packaging. We prototyped the LiDAR scanning module by connecting the LiDAR Lite to servos and a microcontroller, which allows scanning the surroundings and creating a 3D plot of point-cloud data.

7. Future work: UGV design

A major benefit of the proposed fleet configuration is the teaming between UAVs and UGVs. We can combine their different capabilities to maximize fleet performance and efficiency. The next step for fleet design development is the detailed design of the UGV embodiment. Two main issues must be addressed: the off-road capabilities of the UGVs and the interface with the UAVs.

For the off-road capabilities, body stiffness, ability to move through rubble and obstacles in different shapes, and mobility under different terrains must be designed to achieve the required performance.

Another important element in teaming is the replacement of the UAV batteries by the UGVs. For that purpose, the battery casing for the UAV and the replacement mechanism in the UGV must be designed together to make the replacement as easy and quick as possible. The initial functional design we develop at the conceptual design stage features the battery to slide through rails mounted in the casing. In that way, the UGV can quickly remove the old battery and push the new one in. This initial idea might be iterated based on the replacement mechanism solutions developed for the UGV. For instance, we assume

图：原文第 10 页，展示从低保真探索原型到二代聚碳酸酯 3D 打印样机。

贡献与局限

主要贡献不是某一台无人机，而是把“车队运行仿真”插入模块化设计决策中，并明确指出自主性与模块化存在协同效应。

局限也很明显：UGV 尚未完成实体设计；自主换电只做到接口与概念层；车队仿真参数、控制策略和现实救灾不确定性的披露有限；最终原型主要验证机械接口和封装可行性，并未完成完整任务级实证。

可复用的研究框架

如果用于自己的课题，可以沿着以下闭环展开：

任务场景 → 可量化需求 → 多种模块架构 → 任务级蒙特卡洛仿真 → 加权选型 → 实体设计/有限元
→ 原型试验 → 回写模型

尤其值得复用的是评价层级：模块边界的好坏不应只由“接口少”决定，还要看它让整个系统在任务时间、能源、冗余、维护和调度上付出了什么代价。

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